Murtaza Majid

Data Science Undergraduate

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Summary

Final-year Data Science student at Air University with 3 internships and 9 deployed ML/AI projects spanning computer vision, NLP, RAG pipelines, and tabular ML. Built and shipped production systems serving real users — from a citizen AI assistant (Awam Assist) to a 5-model marketing ML pipeline. Currently building a production MLOps pipeline on AWS with MLflow, GitHub Actions CI/CD, and Evidently AI drift monitoring.

Skills

Languages: Python, SQL
ML & DL: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, XGBoost, LSTM, ARIMA, CNNs, NLP, OpenCV, MediaPipe, YOLOv11, Roboflow, SHAP
LLM & RAG: LangChain, ChromaDB, HuggingFace, Groq API, LLaMA 3.3-70B, LangGraph, Prompt Engineering, RLHF, RAG Pipelines
Tools & Platforms: FastAPI, Streamlit, Docker, MLflow, GitHub Actions, Supabase, MySQL, Vercel, Railway, Power BI

Experience

Computer Vision & AI Intern **Dec 2025 – Feb 2026**
Scoutrex, Islamabad *PyTorch, OpenCV, MediaPipe, Roboflow*

- Optimized **PyTorch** models for cricket video analysis under motion blur and varied lighting; implemented **MediaPipe** pose extraction for biomechanical analysis and bowler speed estimation via FPS-normalized displacement.
- Engineered an action phase segmentation pipeline classifying bowling stages (run-up to follow-through) using temporal motion tracking and multi-angle camera synchronization.

LLM Task Trainer **Nov 2025 – Dec 2025**
Turing *Prompt Engineering, RLHF, Tool Use*

- Engineered **system/user prompts** and multi-step task chains to optimize instruction following and integrate external API tools (Search, News, Yelp) using temporal logic.
- Mitigated **hallucinations** and improved model reasoning through rigorous **RLHF** evaluation, ensuring high-accuracy retrieval and context-aware response generation.

Data Science Intern **Jul 2025 – Sep 2025**
ITSOLERA Pvt. Ltd., Islamabad *Python, Pandas, Scikit-Learn, Streamlit*

- Improved time-series forecast accuracy by **15%** using LSTM, ARIMA, XGBoost, and Random Forest models on business metrics.
- Built interactive **Streamlit dashboards** to visualize trends, KPIs, and model predictions for non-technical stakeholders.

Projects

Awam Assist | *LangChain, ChromaDB, HuggingFace, Groq LLaMA 3.3-70B, FastAPI, React*

- Built a **production RAG chatbot** for Pakistani citizens covering 15 government services (NADRA, Passport, Zakat, BISP, FBR, and more), deployed live on Vercel and Railway with bilingual support (English and Roman Urdu) and browser-native **Speech-to-Text** for users with low digital literacy.
- Engineered the full pipeline from document ingestion to deployment: HuggingFace embeddings, ChromaDB vector store, LangChain LCEL retrieval chain, and a FastAPI backend serving real-time answers grounded in verified government sources.

Campbell's AI Marketing Hub | *XGBoost, Scikit-Learn, FastAPI, Supabase, Groq, React*

- Built a 5-model ML pipeline on 12,545 transactions: KMeans RFM segmentation, two-tier churn prediction (**ROC-AUC 0.84**), and ABSA (**F1 0.75**); used time-based split to eliminate data leakage.
- Deployed **16-endpoint FastAPI** on Railway with Supabase; generated personalized SMS/email/app messages via **Groq LLaMA 3.3-70B** using 7-feature behavioral profiling across **2,041 customers**.

Gmail Email Classification Pipeline | *Scikit-Learn, TensorFlow, FastAPI, Supabase, Render*

- Built end-to-end NLP pipeline classifying **5,000+ Gmail emails** into five categories using a 13-step preprocessing pipeline and TF-IDF; achieved **81.0% accuracy** with LSTM, outperforming Naive Bayes (72.6%) and Linear SVM (79.5%).
- Deployed FastAPI on Render with Gmail OAuth and Supabase; delivered live Vercel app with model selection, batch inference, and per-class confidence scoring.

Awards & Certifications

Winner – Air MLX Hackathon 2025: Achieved **RMSE of 0.13** in parking violation risk prediction using multi-table SQL extraction and PCA-based feature reduction.

Certifications: Intermediate Python (DataCamp); Hadoop and Spark (Coursera); British Airways Data Science Simulation (Forage).

Education

Bachelor of Science in Data Science **Expected: June 2026**
Air University, Islamabad